
BCA EXAMINATION NOTES

Course Code: CA-126

CA-126 Introduction to Graphics Design

Exam Pattern:

Q1. Attempt Any Six – 6 marks

Q2. Attempt Any Three – 6 marks

Q3. Attempt Any Two – 6 marks

Q4A. Attempt Any One – 4 marks | Q4B. Compulsory – 2 marks

Q5. Attempt Any One – 6 marks

UNIT 1: Introduction to Graphics Design

1.1 Introduction to Graphics Design

Graphics design is the art and practice of planning and projecting ideas and experiences using visual and textual content. It involves creating visual communication using typography, photography, iconography, and illustration. The purpose is to convey information or ideas visually.

1.2 Types of Graphics

- **Raster/Bitmap Graphics:** Made of pixels. Resolution-dependent (quality decreases on enlargement). Formats: JPEG, PNG, BMP, GIF. Used in photographs.
- **Vector Graphics:** Made of mathematical paths. Resolution-independent (can scale without quality loss). Formats: SVG, AI, EPS. Used in logos, icons.
- **3D Graphics:** Three-dimensional representations rendered using software. Used in games, movies, architecture.
- **Infographics:** Visual representations of data or information combining graphics with text.
- **Motion Graphics:** Animated graphic design – moving images, animations, video effects.

1.3 Uses of Graphics

- Advertising and marketing campaigns
- Website and app user interface design
- Brand identity (logos, packaging)
- Educational materials and textbooks
- Publication design (books, magazines, newspapers)
- Film and television production
- Social media content creation
- Game design and development

1.4 Applications of Graphics Design

- Adobe Photoshop: Raster image editing
- Adobe Illustrator: Vector graphics design
- CorelDraw: Vector illustration and page layout
- Adobe InDesign: Professional publishing layout
- GIMP: Free open-source image editing
- Canva: Web-based graphic design tool
- Blender: 3D modeling and animation (free)

1.5 Different Image Formats

JPEG/JPG: Lossy compression. Best for photographs. Small file size. Does not support transparency.

PNG: Lossless compression. Supports transparency (alpha channel). Best for logos, icons.

GIF: Supports animation. Limited to 256 colors. Used for simple animations on web.

BMP: Bitmap. No compression. Large file size. High quality. Mainly for Windows.

TIFF: High quality, lossless. Used in professional printing. Large files.

SVG: Scalable Vector Graphics. XML-based. Resolution-independent. Good for web icons.

PSD: Photoshop native format. Preserves layers. Not for web distribution.

AI: Adobe Illustrator native format. Vector-based.

EPS: Encapsulated PostScript. Vector format for high-quality print.

WebP: Modern web format by Google. Better compression than JPEG/PNG.

UNIT 2: Introduction to Animation

2.1 What is Animation?

Animation is the process of creating the illusion of motion by displaying a series of still images (frames) in rapid succession. Each frame shows a small change from the previous one. The human eye perceives this sequence as continuous movement due to persistence of vision.

2.2 Principles of Animation

The 12 principles of animation (Frank Thomas & Ollie Johnston):

- Squash and Stretch: Objects deform to show weight and flexibility
- Anticipation: Prepare the audience for an action
- Staging: Present an idea clearly
- Straight Ahead and Pose to Pose: Different approaches to drawing frames
- Follow Through and Overlapping Action: Different parts continue after main action stops
- Slow In and Slow Out: More frames at start/end of movement
- Arc: Natural movements follow arc paths
- Secondary Action: Additional movement supporting main action
- Timing: Speed of animation conveys emotion and weight
- Exaggeration: Amplify actions for dramatic effect
- Solid Drawing: 3D thinking in 2D drawing
- Appeal: Characters should be interesting and appealing

2.3 Types of Animation and Animation Techniques

- Traditional (Hand-Drawn) Animation: Drawing each frame by hand on paper. e.g., early Disney films.
- 2D Computer Animation: Digital drawing using software. e.g., Flash animation, Adobe Animate.
- 3D Computer Animation: Creating 3D models and animating them. e.g., Pixar films.
- Stop Motion Animation: Photographing physical objects moved in small increments. e.g., Claymation.
- Motion Capture (MoCap): Recording real actor movements applied to digital characters.
- Cutout Animation: Moving flat characters made from paper/cardboard. e.g., South Park.
- Rotoscoping: Tracing over live-action footage frame by frame.
- Morphing: Smooth transformation from one shape/image to another.

2.4 Introduction to Chroma Shoot (Chroma Key)

Chroma key (green screen or blue screen) is a visual effects technique where a solid-colored background (usually green or blue) is filmed and later replaced with another background in post-production.

Process: Actor filmed against green screen → Green color is keyed out (removed) by software → Replaced with any desired background.

Used in: Weather broadcasts, movies (virtual environments), live streaming overlays, YouTube production.

Software: Adobe After Effects, OBS Studio, DaVinci Resolve.

2.5 Use of Animation in Advertisements

- Explain complex products through animated demos

- Create brand mascots and characters
- Cost-effective compared to live-action shoots
- Can create impossible scenarios (fantasy, sci-fi)
- Animated logos and transitions in TV commercials
- Social media motion graphics for engagement
- Explainer videos for startups and apps

2.6 Concept of 2D and 3D Animation

2D ANIMATION: Two-dimensional space (X and Y axes only). Flat characters moving on a plane. Simpler to create, less resource-intensive. Tools: Adobe Animate, Toon Boom, After Effects. Examples: Tom & Jerry, SpongeBob.

3D ANIMATION: Three-dimensional space (X, Y, and Z axes). Objects have depth, realistic lighting and shadows. More complex and resource-intensive. Tools: Blender, Maya, Cinema 4D, 3ds Max. Examples: Toy Story, Avatar, most video games.

UNIT 3: Working with Photoshop CS3

3.1 Photoshop Workspace

Adobe Photoshop CS3 workspace components:

- Menu Bar: File, Edit, Image, Layer, Select, Filter, View, Window, Help
- Options Bar: Tool-specific options for the selected tool
- Toolbox/Tools Panel: Collection of all tools on the left
- Canvas/Image Area: The main working area
- Layers Panel: Manage image layers
- History Panel: Undo/redo actions
- Properties/Info Panel: Shows details about selection/tool

3.2 Features of Photoshop

- Layer-based editing – non-destructive editing
- Advanced selection tools
- Filters and effects library
- Color correction and adjustment tools
- Smart Objects – non-destructive transformations
- Masks – hide/reveal parts without deleting
- Text effects and typography tools
- Batch processing – automate repetitive tasks

3.3 Photoshop in Different Sectors

- Photography: Retouching, color correction, compositing
- Advertising: Creating promotional banners, ads
- Publishing: Book covers, magazine layouts
- Web Design: UI mockups, web graphics
- Film: Movie posters, visual effects
- Fashion: Photo retouching, clothing visualization
- Medical: Analyzing medical imaging

3.4 Basic Tools

Select Tools: Rectangular Marquee, Elliptical Marquee – select regular shapes

Move Tool (V): Move layers and selections

Zoom Tool (Z): Zoom in/out of the canvas

Hand Tool (H): Pan/scroll the canvas

Shape Tools: Rectangle, Ellipse, Polygon, Custom shapes

Brush Tool (B): Paint on canvas with customizable brush

Fill Tool/Paint Bucket: Fill area with foreground color

Text Tool (T): Add text to the image

Eyedropper Tool: Sample/pick colors from image

Crop Tool (C): Crop the image to selected area

3.5 Advanced Tools

Marquee Tools: Rectangular/Elliptical – precise selections

Lasso Tool: Freehand selection – regular, Polygonal, Magnetic

Magic Wand Tool: Selects based on color similarity (tolerance setting)

Blur Tool: Softens/blurs areas of the image

Sharpen Tool: Increases clarity/sharpness of edges

Smudge Tool: Pushes and smears pixels, like wet paint

Dodge Tool: Lightens areas (like photographic dodge)

Burn Tool: Darkens areas (like photographic burn)

Sponge Tool: Changes saturation of an area (saturate/desaturate)

Clone Stamp Tool: Copies pixels from one area to another – used for retouching

3.6 Transformation and Retouching

Edit > Free Transform (Ctrl+T) allows:

- Scaling: Resize the layer
- Rotation: Rotate the layer around center
- Skew: Slant horizontally or vertically
- Perspective: Apply perspective distortion
- Warp: Deform into custom shape
- Distort: Freely move corner handles

Image > Image Size: Change pixel dimensions and resolution

Image > Canvas Size: Extend/reduce canvas without changing image

Image > Crop: Remove outer areas

3.7 Adjustments

Levels (Ctrl+L): Adjust shadows, midtones, highlights using histogram

Curves (Ctrl+M): Fine-tune tonal range with custom curve

Color Balance: Adjust relative mix of colors for shadows/midtones/highlights

Exposure: Adjust overall exposure, offset, and gamma

Vibrance: Intelligently boost muted colors without over-saturating

Hue/Saturation: Shift hue, adjust saturation and lightness

Brightness/Contrast: Simple overall adjustment

Photo Filter: Apply a colored filter effect to image

3.8 Working with Text

Text Tool (T) adds text to images. Types: Horizontal Type Tool, Vertical Type Tool, Type Mask Tool.

Editing Text: Double-click text layer → Cursor appears → Edit text

Formatting: Change font family, size, style (bold/italic), color, kerning, leading, tracking

Text Effects: Drop shadow, outer glow, bevel and emboss – via Layer Styles (fx button)

Wrap Text: Not native in PS; use Warp Text option or text on a path

UNIT 4: Working with CorelDraw

4.1 Introduction and Interface

CorelDraw is a vector graphics editor developed by Corel Corporation. Used for logo design, illustrations, page layout, brochures, and technical drawings.

Interface components:

- Menu Bar: File, Edit, View, Layout, Arrange, Effects, Bitmaps, Text, Tools, Window, Help
- Toolbox: Drawing and editing tools on the left
- Property Bar: Context-sensitive options for selected tool
- Rulers: Measure and align objects
- Drawing Page: Main working area
- Color Palette: Click to fill, right-click to outline
- Status Bar: Shows object info, position, size
- Docker Panels: Object properties, layers, etc.

4.2 Drawing: Lines, Shapes, Objects, Tables

- Straight Lines: Freehand tool, Bezier tool
- Curves: Pen tool, B-Spline tool
- Shapes: Rectangle (F6), Ellipse (F7), Polygon, Star
- Objects: Grouped objects, symbols, smart drawing
- Tables: Table tool for grid/tabular layouts
- Templates: Pre-designed layouts to start quickly

4.3 Import/Export

Import: File > Import (Ctrl+I) – Brings external files (photos, logos) into CorelDraw

Corel Connect: Asset library browser for finding and using graphics/fonts/templates

Export: File > Export (Ctrl+E) – Save in different formats: PNG, JPEG, PDF, SVG, EPS, AI

4.4 Manipulation

Pick Tool: Select, move, scale, rotate, skew objects

Selection: Click to select; Shift+click for multiple; rubber-band drag for all in area

Moving: Drag with Pick tool or use arrow keys

Sizing: Drag handles; hold Ctrl to maintain proportion

Mirroring: Horizontal/vertical flip from Property Bar

Rotating: Click selected object twice – rotation handles appear; drag corner handles

Skewing: Drag side handles in rotation mode

Undo/Redo: Ctrl+Z / Ctrl+Y — multiple undo levels

Stacking Order: Arrange > Order: Bring to Front, Send to Back, etc.

4.5 Text Properties in CorelDraw

Artistic Text: Click anywhere with Text tool – for short text. Can apply effects.

Paragraph Text: Drag a text frame – for long text (articles, stories)

Properties: Font, size, bold, italic, underline, alignment, color, spacing

Fit Text to Path: Make text flow along a curve/shape

Text Wrap: Make surrounding objects wrap around text frame

DEFINITIONS (2-Mark Ready)

Raster Graphics: Raster graphics are image files made of pixels (picture elements) arranged in a grid, where quality degrades when scaled up. Examples: JPEG, PNG, BMP.

Vector Graphics: Vector graphics use mathematical paths and formulas to define images, making them resolution-independent and infinitely scalable without quality loss. Examples: SVG, AI.

Animation: Animation is the technique of creating an illusion of motion by displaying a series of still frames in rapid succession, exploiting the persistence of vision.

Chroma Key: Chroma key is a visual effects technique where a solid-color background (green/blue screen) is removed and replaced with another image or scene in post-production.

Layer (Photoshop): A layer is an independent level of an image in Photoshop that can be edited separately without affecting other layers, enabling non-destructive editing.

Clone Stamp: The Clone Stamp tool in Photoshop copies pixels from one area of an image to another, used for removing blemishes, objects, or retouching photos.

Levels Adjustment: Levels is a Photoshop adjustment that controls the tonal range of an image by adjusting shadows, midtones, and highlights using a histogram.

CorelDraw: CorelDraw is a professional vector graphics design application used for logo creation, illustrations, page layouts, and technical drawings.

JPEG: JPEG (Joint Photographic Experts Group) is a lossy image compression format ideal for photographs, offering small file sizes at the cost of some quality.

PNG: PNG (Portable Network Graphics) is a lossless image format that supports transparency (alpha channel), commonly used for logos and web graphics.

20 MOST IMPORTANT QUESTIONS WITH ANSWERS

Q1. What is Graphics Design? State its uses and applications. [6M]

Graphics design is the art of creating visual content to communicate messages using typography, color, images, and layout.

Uses:

1. Advertising and marketing (banners, posters)
2. Website and mobile app design

3. Brand identity (logos, business cards)
4. Publication design (books, magazines)
5. Packaging design for products
6. Social media content
7. Educational materials

Applications/Software:

1. Adobe Photoshop – photo editing
2. Adobe Illustrator – vector design
3. CorelDraw – vector + layout
4. Adobe InDesign – publishing
5. Canva – online design
6. GIMP – free photo editor
7. Blender – 3D design and animation

Q2. Explain types of graphics with examples. [4M]

1. RASTER/BITMAP GRAPHICS:

Made of pixels. Quality degrades when enlarged.

Formats: JPEG, PNG, GIF, BMP

Used for: Photographs, web images

2. VECTOR GRAPHICS:

Made of mathematical paths. Scale without quality loss.

Formats: SVG, AI, EPS, CDR

Used for: Logos, icons, illustrations

3. 3D GRAPHICS:

Three-dimensional. Used in games, movies.

Software: Blender, Maya, 3ds Max

4. INFOGRAPHICS:

Combination of data + visual graphics.

Used to present complex info simply.

5. MOTION GRAPHICS:

Animated graphic elements.

Used in title sequences, ads, presentations.

Q3. Explain different image formats with their uses. [6M]

1. JPEG: Lossy compression. Best for photos. Small size. No transparency. Web photos.
2. PNG: Lossless. Supports transparency. Best for logos, icons, screenshots.
3. GIF: Supports animation. Only 256 colors. Web animations, simple graphics.
4. BMP: No compression. Very high quality. Large file size. Windows native.
5. TIFF: High quality, lossless. Used in professional printing. Large files.
6. SVG: Scalable vector. XML-based. Perfect for web icons, logos.
7. PSD: Photoshop native. Preserves layers. Not for distribution.
8. AI: Adobe Illustrator native. Vector.
9. EPS: Vector. Used for high-quality print.
10. WebP: Modern web format. Better quality than JPEG at same size.

Q4. What is animation? Explain its types. [6M]

Animation is the technique of creating illusion of movement by displaying a series of still images (frames) in rapid succession.

Types:

1. TRADITIONAL/HAND-DRAWN: Each frame drawn by hand. e.g., early Disney, Tom & Jerry.
2. 2D COMPUTER ANIMATION: Digital 2D drawing and movement. e.g., Flash animations.
3. 3D COMPUTER ANIMATION: 3D models animated in software. e.g., Toy Story, Avatar.
4. STOP MOTION: Physical objects moved slightly between photographs. e.g., Claymation.
5. MOTION CAPTURE: Real actor movements digitally applied to characters.
6. CUTOUT ANIMATION: Flat characters cut from paper/card moved and photographed.
7. ROTOSCOPING: Tracing over live-action footage frame by frame.
8. MORPHING: Smooth transformation from one shape/image to another.

Q5. Explain principles of animation. [6M]

The 12 principles of animation (by Disney animators Frank Thomas and Ollie Johnston):

1. Squash and Stretch: Show weight and flexibility of objects.
2. Anticipation: Prepare viewer for main action.
3. Staging: Present idea clearly – like theatrical staging.
4. Straight Ahead/Pose to Pose: Two drawing approaches.
5. Follow Through: Parts continue moving after main action stops.
6. Slow In and Slow Out: More frames at start/end for natural movement.
7. Arc: Natural movements follow curved paths.
8. Secondary Action: Supporting movement that adds to main action.
9. Timing: Speed of action conveys weight and personality.
10. Exaggeration: Amplify for dramatic effect.
11. Solid Drawing: 3D thinking in 2D drawing.
12. Appeal: Characters must be visually interesting and engaging.

Q6. What is chroma key (green screen)? Explain its uses. [4M]

Chroma key (also called green screen or blue screen) is a visual effects technique where actors/subjects are filmed in front of a solid-color background (usually green or blue), which is then digitally removed and replaced with any desired background or scene in post-production.

Process:

1. Set up green/blue screen background
2. Film subject against the screen with good lighting
3. In editing software, 'key out' the green/blue color
4. Replace with desired background footage or image

Uses:

1. Weather broadcasting (maps behind presenter)
2. Movies – virtual environments (Avengers, Avatar)
3. YouTube videos – virtual backgrounds
4. Live streaming (OBS Studio)
5. News reporting – map/graphic overlays

Software: Adobe After Effects, OBS, DaVinci Resolve, Final Cut Pro

Q7. Explain the Photoshop workspace. [4M]

Adobe Photoshop CS3 workspace includes:

1. MENU BAR: Top bar with menus – File, Edit, Image, Layer, Select, Filter, View, Window, Help.
2. OPTIONS BAR: Below menu bar. Shows options specific to the currently selected tool.
3. TOOLBOX (Tools Panel): Left side panel with all editing tools – selection, drawing, retouching, etc.
4. CANVAS/IMAGE AREA: Central area where the image is displayed and edited.
5. LAYERS PANEL: Right side. Shows all layers; allows layer management.
6. HISTORY PANEL: Shows list of recent actions; allows undo multiple steps.
7. COLOR SWATCHES PANEL: Quick access to colors.
8. NAVIGATOR: Small preview of entire image for navigation.

Q8. Explain Photoshop selection tools. [4M]

Selection tools allow isolating specific areas of an image for editing:

1. RECTANGULAR MARQUEE TOOL: Creates rectangular/square selections. Hold Shift for square.
2. ELLIPTICAL MARQUEE TOOL: Creates oval/circular selections. Hold Shift for perfect circle.
3. LASSO TOOL: Freehand selection – draw any irregular shape by holding mouse.
4. POLYGONAL LASSO TOOL: Click to create straight-line segments forming irregular selection.
5. MAGNETIC LASSO TOOL: Snaps to edges automatically – great for selecting objects with distinct edges.
6. MAGIC WAND TOOL: Selects pixels of similar color. Tolerance setting controls color range.
7. QUICK SELECTION TOOL: Brush-based tool – paint over area to select it automatically.

Modifiers: Shift+drag (add to selection), Alt+drag (subtract from selection)

Q9. Explain transformation options in Photoshop. [4M]

Transformation in Photoshop: Edit > Free Transform (Ctrl+T)

Transformation Handle appears around selected layer/object.

Transformation Options:

1. SCALING: Drag corner handle to resize. Hold Shift to maintain proportion.
2. ROTATION: Move cursor slightly outside corner handle → rotation cursor appears → drag to rotate.
3. SKEW: Right-click > Skew → Drag side handles to slant horizontally or vertically.
4. PERSPECTIVE: Right-click > Perspective → Drag corner to create perspective effect.
5. WARP: Right-click > Warp → Deform into custom curved shape using grid points.
6. DISTORT: Right-click > Distort → Move corner handles independently.
7. FLIP: Edit > Transform > Flip Horizontal / Flip Vertical.

8. CROP: Crop Tool (C) → Draw selection → Press Enter.

Q10. Explain image adjustments in Photoshop. [6M]

Image > Adjustments provide color and tonal corrections:

1. LEVELS (Ctrl+L):

Adjust shadows (black point), midtones (gamma), highlights (white point).

Use histogram to optimize tonal range.

2. CURVES (Ctrl+M):

Precise tonal control using a customizable curve.

Drag curve up (brighten) or down (darken) for specific tones.

3. COLOR BALANCE:

Adjust mix of Cyan/Red, Magenta/Green, Yellow/Blue.

Separately for Shadows, Midtones, Highlights.

4. EXPOSURE:

Simulate camera exposure. Adjust Exposure, Offset (blacks), Gamma.

5. VIBRANCE:

Smartly boosts muted colors without over-saturating skin tones.

6. HUE/SATURATION (Ctrl+U):

Shift colors (Hue), adjust color intensity (Saturation), lightness.

7. PHOTO FILTER:

Apply color filter effect (warming, cooling) to the image.

Q11. Explain CorelDraw interface. [4M]

CorelDraw is a professional vector graphics application. Its interface:

1. MENU BAR: File, Edit, View, Layout, Arrange, Effects, Bitmaps, Text, Tools, Window, Help.

2. STANDARD TOOLBAR: Common operations – New, Open, Save, Cut, Copy, Paste, Undo, Redo.

3. PROPERTY BAR: Context-sensitive options that change based on selected tool.

4. TOOLBOX: Left side – Drawing tools, text, shaping, zooming etc.

5. DRAWING PAGE: White rectangle = printable area. Gray area = workspace.

6. RULERS: Top and left rulers for measurement and alignment.

7. COLOR PALETTE: Right side strip. Left-click = fill object; Right-click = outline.

8. STATUS BAR: Bottom – shows selected object info, position, size.

Q12. Explain drawing tools in CorelDraw. [4M]

CorelDraw offers comprehensive drawing tools:

1. PICK TOOL (F1): Select, move, resize, rotate objects.
2. FREEHAND TOOL (F5): Draw lines and curves by dragging.
3. BEZIER/PEN TOOL: Draw precise curves using control points.
4. RECTANGLE TOOL (F6): Draw rectangles/squares.
5. ELLIPSE TOOL (F7): Draw ellipses/circles.
6. POLYGON TOOL: Draw polygons and stars.
7. SPIRAL TOOL: Draw spiral shapes.
8. TEXT TOOL (F8): Add text to the document.
9. SMART DRAWING TOOL: Recognizes and corrects hand-drawn shapes.
10. TABLE TOOL: Create tables in the document.

Templates: File > New from Template – pre-designed layouts for brochures, business cards, etc.

Q13. Explain advanced tools in Photoshop. [4M]

Photoshop Advanced Retouching Tools:

1. CLONE STAMP TOOL (S): Alt+click to sample → Paint to copy pixels. Used to remove blemishes/objects.
2. HEALING BRUSH TOOL: Like Clone Stamp but blends texture with surrounding area naturally.
3. PATCH TOOL: Select area → drag to good area → blends automatically.
4. BLUR TOOL: Softens sharp edges or distracting details.
5. SHARPEN TOOL: Increases contrast at edges for more apparent sharpness.
6. SMUDGE TOOL: Pushes pixels as if smearing wet paint.
7. DODGE TOOL (O): Lightens areas (exposure-based).
8. BURN TOOL: Darkens areas.
9. SPONGE TOOL: Saturates or desaturates color in specific areas.
10. MAGIC WAND TOOL: Selects pixels of similar color (tolerance controls range).

Q14. What is the difference between raster and vector graphics? [4M]

RASTER GRAPHICS:

1. Made of pixels (dots of color)
2. Resolution-dependent – quality drops when enlarged
3. Large file size at high resolution
4. Best for photographs and complex images
5. Formats: JPEG, PNG, BMP, GIF
6. Software: Photoshop, GIMP

VECTOR GRAPHICS:

1. Made of mathematical paths, points, curves
2. Resolution-independent – infinitely scalable without quality loss
3. Smaller file size
4. Best for logos, icons, illustrations
5. Formats: SVG, AI, EPS, CDR
6. Software: Illustrator, CorelDraw

Key Difference: Raster loses quality when scaled up; vector does not.

Q15. Explain text tools in Photoshop. [4M]

Photoshop Text Tool (T) features:

1. HORIZONTAL TYPE TOOL: Add text horizontally anywhere on canvas.
2. VERTICAL TYPE TOOL: Add text vertically.
3. HORIZONTAL TYPE MASK TOOL: Creates a selection in the shape of text.
4. VERTICAL TYPE MASK TOOL: Same but vertical.

Adding Text: Select Text tool → Click on canvas → Type → Select Move tool to position.

Formatting:

- Font family: Choose typeface (Arial, Times New Roman)
- Font size: Measured in points
- Style: Bold, Italic, Regular
- Color: Pick from color picker
- Leading: Vertical space between lines
- Kerning/Tracking: Horizontal space between characters
- Alignment: Left, Center, Right, Justify

Text Effects: Double-click layer → Layer Style dialog → Drop shadow, Outer glow, Bevel and Emboss, Stroke

Q16. What is the use of layers in Photoshop? [4M]

Layers are independent stacked sheets in Photoshop, each containing different elements (text, image, shape).

Benefits of Layers:

1. Non-destructive editing – edit one layer without affecting others
2. Easy to rearrange elements by reordering layers
3. Each layer can have its own opacity and blend mode
4. Layer Masks hide/reveal parts without deleting
5. Adjustment Layers apply color corrections non-destructively

6. Text and shape layers remain editable

Layer Operations:

- Create new layer: Layer > New Layer or Ctrl+Shift+N
- Duplicate: Ctrl+J
- Delete: Select + drag to trash icon
- Merge: Ctrl+E (merge selected layers)
- Flatten: Image > Flatten Image (merge all)

Q17. Explain object manipulation in CorelDraw. [4M]

CorelDraw provides comprehensive object manipulation:

1. SELECTION: Click with Pick tool; Shift+click for multiple; rubber-band drag.
2. MOVING: Drag selected object; use arrow keys for precise movement.
3. SIZING: Drag corner/side handles; enter exact values in Property Bar.
4. MIRRORING: Click Mirror Horizontal or Mirror Vertical in Property Bar.
5. ROTATING: Click selected object a second time → rotation handles appear → drag corner.
6. SKEWING: In rotation mode, drag side handles to skew.
7. UNDO: Ctrl+Z (undo); Ctrl+Y (redo) – multiple levels.
8. STACKING ORDER: Arrange > Order:
 - Bring to Front (Shift+PgUp)
 - Send to Back (Shift+PgDown)
 - Bring Forward (PgUp)
 - Send Backward (PgDown)

Q18. What is the 2D vs 3D animation concept? [4M]

2D ANIMATION:

1. Two-dimensional space – X and Y axes only
2. Flat characters and backgrounds
3. Simpler to create, faster, less resource-intensive
4. Frame-by-frame or tweening animation
5. Tools: Adobe Animate, Toon Boom, After Effects
6. Examples: Tom & Jerry, SpongeBob, Aladdin
7. Used in: TV cartoons, educational videos, web animations

3D ANIMATION:

1. Three-dimensional space – X, Y, and Z axes
2. Objects have depth, volume, realistic lighting and shadows
3. Complex workflow: modelling → rigging → animating → rendering
4. More time and resource-intensive
5. Tools: Blender, Maya, 3ds Max, Cinema 4D
6. Examples: Toy Story, Avatar, most video games

7. Used in: Feature films, games, architectural visualization

Q19. What is CorelDraw used for? Explain import and export. [4M]

CorelDraw is a professional vector graphics application used for:

- Logo design and brand identity
- Illustrations and technical drawings
- Brochure and flyer layout
- Business card, letterhead design
- Book and magazine layout
- Banner and poster design

IMPORT: File > Import (Ctrl+I)

Brings external files into the CorelDraw document.

Can import: JPEG, PNG, BMP, PSD, AI, SVG, PDF photos etc.

Corel Connect: Asset browser to find clipart, fonts, templates.

EXPORT: File > Export (Ctrl+E)

Saves CorelDraw content in various formats:

- PNG: For web (transparent background)
- JPEG: For web photos
- PDF: For printing
- SVG: For web vectors
- EPS/AI: For professional printing/other software

Q20. Explain Photoshop's uses in different sectors. [2M]

Photoshop is used across many industries:

1. Photography: Photo retouching, color correction, HDR, compositing
2. Advertising: Creating banners, posters, promotional materials
3. Web Design: Designing UI mockups, web graphics, icons
4. Film/TV: Movie posters, visual effects, matte painting
5. Fashion: Retouching model photos, virtual clothing design
6. Publishing: Book covers, magazine page design
7. Medical: Enhancing and annotating medical imaging
8. Architecture: Rendering architectural visualizations
9. Game Design: Creating textures, concept art, sprites
10. Social Media: Designing posts, thumbnails, profile images